

New Techniques for Analyzing Block Cipher Differentials

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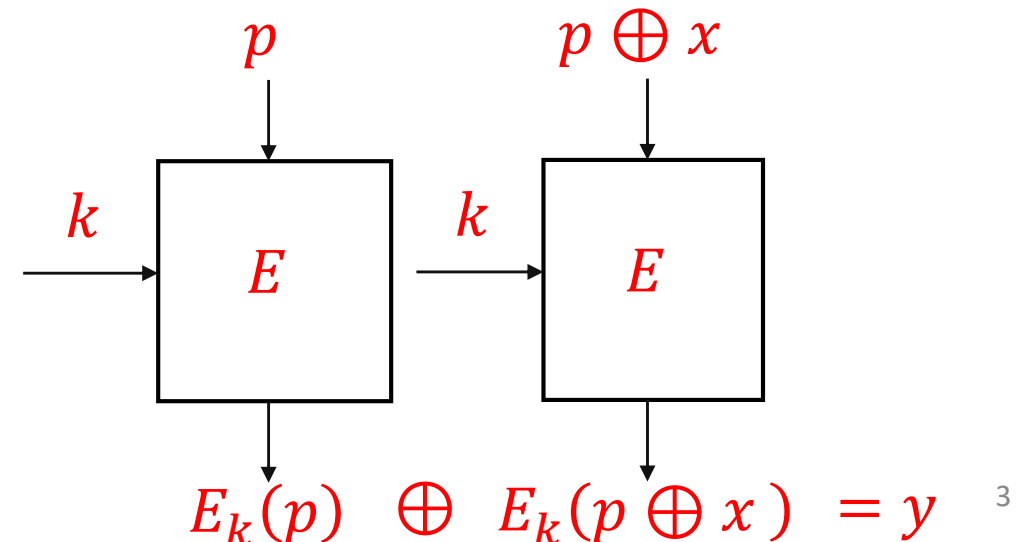
Based on <https://eprint.iacr.org/2025/1326>

Differential Cryptanalysis (Biham and Shamir '90)

- **Differential Cryptanalysis** is one of the most powerful attacks on block ciphers
- Consider block cipher $E: \{0,1\}^k \times \{0,1\}^b \rightarrow \{0,1\}^b$
- A (nonzero) **differential** for E is a pair of (nonzero) input difference $x \in \{0,1\}^b$ and (nonzero) output difference $y \in \{0,1\}^b$
- Probability of **differential** $\Pr[x \rightarrow_E y]$:

$$\Pr_{k \in \{0,1\}^k, p \in \{0,1\}^b} [E_k(p) \oplus E_k(p \oplus x) = y]$$

- Note: $\Pr[\mathbf{0} \rightarrow_E \mathbf{0}] = 1$



Differential Cryptanalysis (Biham and Shamir '90)

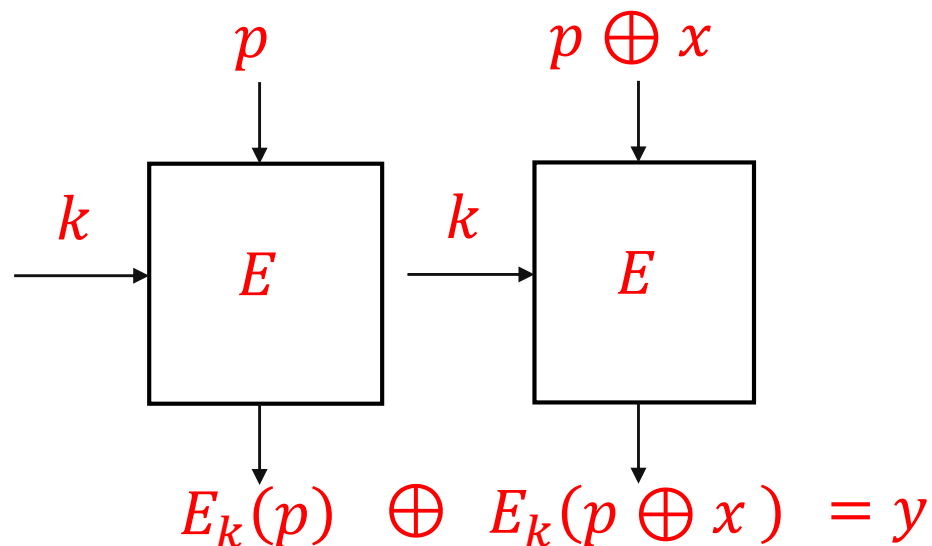
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$$\Pr_{k \in \{0,1\}^k, p \in \{0,1\}^b} [E_k(p) \oplus E_k(p \oplus x) = y]$$

- Note: $\Pr[\mathbf{0} \rightarrow_E \mathbf{0}] = 1$
- Define difference distribution table (**DDT**) matrix by $D_E[y, x] = \Pr[x \rightarrow_E y]$
 - $D_E \in \mathbb{R}^{2^b \times 2^b}$
 - For minor technical reasons, more convenient to index the columns by input diffs₄

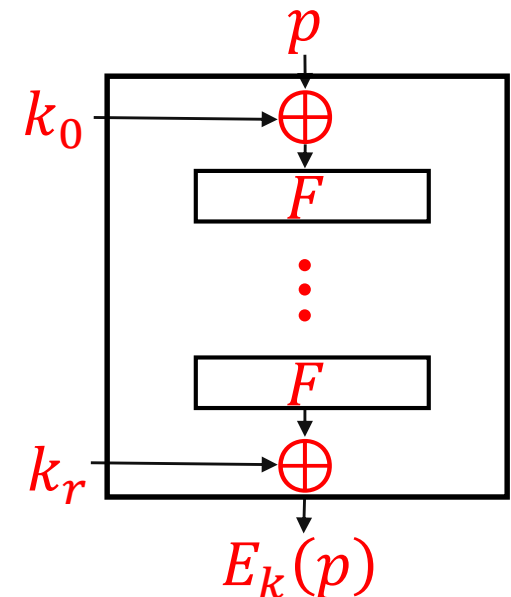
Differential Cryptanalysis (Biham and Shamir '90)

- A (classical) differential **distinguisher** consists of nonzero (x, y) such that $D_E[y, x] = \Pr[x \rightarrow_E y] \gg \frac{1}{2^b - 1}$ (probability in a uniform keyed permutation)
- To prove that E **resists** differential cryptanalysis (at least) show that for all nonzero (x, y) we have $\Pr[x \rightarrow_E y] \approx \frac{1}{2^b - 1}$
- Modern block ciphers are **complex** and estimating $\Pr[x \rightarrow_E y]$ is **infeasible**



Markov Ciphers (Lai, Massey, and Murphy '91)

- Consider **key-alternating ciphers**:
- Given **public round permutation** $F: \{0,1\}^b \rightarrow \{0,1\}^b$, define:
- $E_k(p) = F(\dots (F(p \oplus k_0) \dots) \oplus k_r$, where $k_0, \dots, k_r$ derived from k using **key-schedule algorithm**
- **Standard simplification**: $k_0, \dots, k_r \in \{0,1\}^\kappa$ chosen independently, uniformly
- E is a **Markov cipher**
- Motivation:
 - Simplest model for analysis
 - Most (differential) attacks make such a simplification



Differential Characteristics

- Estimating differential probability still **challenging** even for Markov ciphers
- Compromise: consider **differential characteristics** which specify how input difference propagates to output difference through internal cipher operations
- Goal: prove every differential characteristic has very low probability
- Not difficult for a well-designed cipher

Example: AES

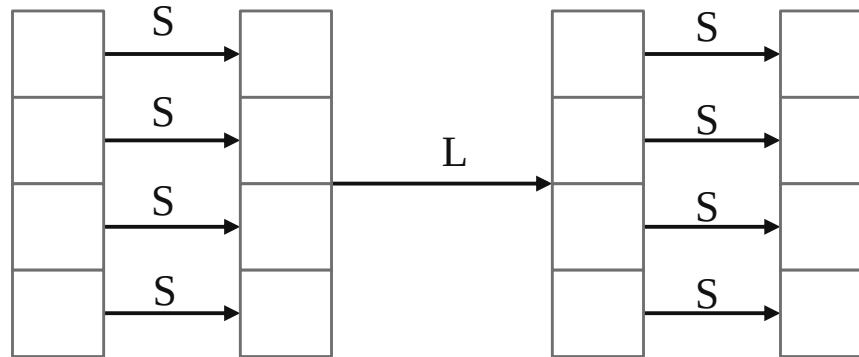
- Consider the **AES** key-alternating cipher:
 - Block length $b = 128$, number of rounds $r=10, 12$ or 14
- Show using “**wide trail strategy**” that every differential characteristic in 4-round AES has probability $\leq 2^{-6 \cdot 25} = 2^{-150}$
- Main non-linear AES component:
- (Keyed) **Sbox function** $S: \{0,1\}^8 \times \{0,1\}^8 \rightarrow \{0,1\}^8$
- For every nonzero (x, y) , $\Pr[x \rightarrow_s y] \leq 2^{-6}$
- Assuming independent round-key, sufficient to show every 4-round characteristic has at least **25 active** (non-zero) Sboxes

Super-Sbox Differential Characteristics

- First analyze characteristics for AES keyed **super-Sbox** function

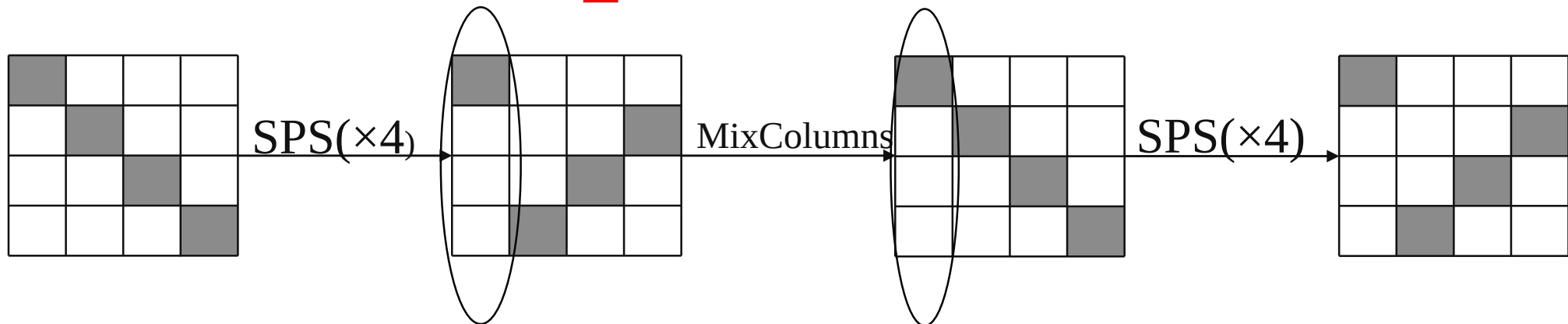
$SPS: \{0,1\}^{64} \times \{0,1\}^{32} \rightarrow \{0,1\}^{32}$:

- 1. Apply keyed Sbox function $S: \{0,1\}^8 \times \{0,1\}^8 \rightarrow \{0,1\}^8$ to each byte
- 2. Apply linear permutation $L: \{0,1\}^{32} \rightarrow \{0,1\}^{32}$
 - Part of MixColumns, implemented by MDS matrix **branch number 5** (over 8-bit words):
 - $wt(x) + wt(L(x)) \geq 5$
- 3. Apply keyed Sbox function $S: \{0,1\}^8 \times \{0,1\}^8 \rightarrow \{0,1\}^8$ to each byte
- By MDS property: any non-zero characteristic in super-Sbox has ≥ 5 **active Sboxes**



4-Round AES Characteristics (No Final MixColumns)

- Arrange the 16-byte state in a 4×4 matrix
- 1. Apply keyed super-Sbox $SPS: \{0,1\}^{64} \times \{0,1\}^{32} \rightarrow \{0,1\}^{32}$ to each of 4 diagonals and place each result in an anti-diagonal
- 2. MixColumns: apply a linear function to each column
 - Branch number 5 over super-Sboxes (32-bit words)
- 3. Repeat 1. : apply super-Sbox to each of 4 diagonals
- By branch number: any non-zero characteristic has ≥ 5 **active super-Sboxes**
- Each of ≥ 5 active super-Sboxes has ≥ 5 active Sboxes
- Every 4-round characteristic has ≥ 25 active Sboxes



Differentials VS. Differential Characteristics

- Differential (x, y) can be composed of **huge number** of characteristics with **tiny probability**
- Overall differential probability can be **large**
- Trivial example: consider “good” 64-bit cipher E
- Any characteristic in E has **tiny probability**, say 2^{-200}
- Assume for all nonzero differentials (x, y) , $\Pr[x \rightarrow_E y] \approx \frac{1}{2^{64}-1} \approx 2^{-64}$
- Define 128-bit cipher G as $G_{k_1, k_2}(p_1 || p_2) = E_{k_1}(p_1) || E_{k_2}(p_2)$
 - Any characteristic in G has **tiny probability**, say 2^{-200}
 - For all differentials $((0, x), (0, y))$ have $\Pr[(0, x) \rightarrow_G (0, y)] \approx 2^{-64} \gg \frac{1}{2^{128}-1}$
- There are also non-trivial examples

Differential Characteristic Bounds

- Differential characteristics are good for **devising attacks**
 - Characteristic with high probability $\rightarrow$ differential with high probability $\rightarrow$ efficient attack
- Bounding probability of differential characteristics good for **arguing security** against **specific attacks**
 - Using one (few) differential characteristics
- **Insufficient** for arguing security against differential cryptanalysis

AES Differentials

- Estimating differential probability still **challenging** even for Markov ciphers
- For AES, ideally prove for **small number of rounds** r , for any nonzero (x, y) :
$$D_{AES(r)}[y, x] = \Pr[x \rightarrow_{AES(r)} y] \approx \frac{1}{2^{128}-1}$$
 (assuming independent round-keys)
- Best-known result for **4**-round AES by Keliher and Sui [KS'07]:
$$\Pr[x \rightarrow_{AES(4)} y] \leq 2^{-113.09}$$
- Better results **not known** for more than **4** rounds
 - For number of rounds using **in practice**

AES Differentials

- Best-known result for 4-round AES [KS'07]: $\Pr[x \rightarrow_{AES(4)} y] \leq 2^{-113.09}$
- **Proves limited resistance** against classical differential cryptanalysis
- **Proves no resistance** against **impossible differential cryptanalysis**:
 $\Pr[x \rightarrow_{AES(4)} y] = 0$ for a subset of nonzero (x, y)
- However, impossible differential cryptanalysis among **most powerful attacks** on round-reduced AES

New Results

- Main result: Prove for 8-round AES: $\left| \Pr[x \rightarrow_{AES(8)} y] - \frac{1}{2^{128}-1} \right| \leq \frac{1}{50} \cdot 2^{-128}$ for any nonzero (x, y)
 - $\Pr[x \rightarrow_{AES(8)} y] \leq \frac{1}{2^{128}-1} + \frac{1}{50} \cdot 2^{-128}$ (resists classical differential cryptanalysis)
 - $\Pr[x \rightarrow_{AES(8)} y] \geq \frac{1}{2^{128}-1} - \frac{1}{50} \cdot 2^{-128}$ (resists impossible differential cryptanalysis)
- Additional results:
 - Improved deviation bound $\left| \Pr[x \rightarrow_{AES(4r)} y] - \frac{1}{2^{128}-1} \right|$ for $r \geq 3$
 - Prove $\Pr[x \rightarrow_{AES(4)} y] \leq 2^{-116.54}$ (improving [KS'07] by factor of ≈ 11)
 - Obtain bounds on 1-norm distance of $AES^{(4r)}$ from pairwise independent permutation
- **Techniques** may be useful for analyzing differentials for **additional block ciphers**

Reformulating Main Result with Norms

- $\left| \Pr[x \rightarrow_{AES(8)} y] - \frac{1}{2^{128}-1} \right| \leq \frac{1}{50} \cdot 2^{-128}$ for any nonzero (x, y)
- For nonzero input diff x , consider output distribution (**vector**) indexed by DDT column x
- Write it as matrix-vector product $D_{AES(8)}^- e_x$ for **unit vector** $e_x \in \mathbb{R}^{2^{128}-1}$
 - $D_{AES(8)}^- \in \mathbb{R}^{(2^{128}-1) \times (2^{128}-1)}$ truncates zero row and column from DDT
- Define U as **uniform** distribution (vector) $\rightarrow U(y) = \frac{1}{2^{128}-1}$ for nonzero y
- Need to prove $\left\| D_{AES(8)}^- e_x - U \right\|_{\infty} \leq \frac{1}{50} \cdot 2^{-128}$ for any nonzero x
 - $\|v\|_{\infty} = \max_i |v[i]|$, $\|v\|_2 = (\sum_i |v[i]|^2)^{\frac{1}{2}}$

Reduction to 4-Round 2-norm Bound

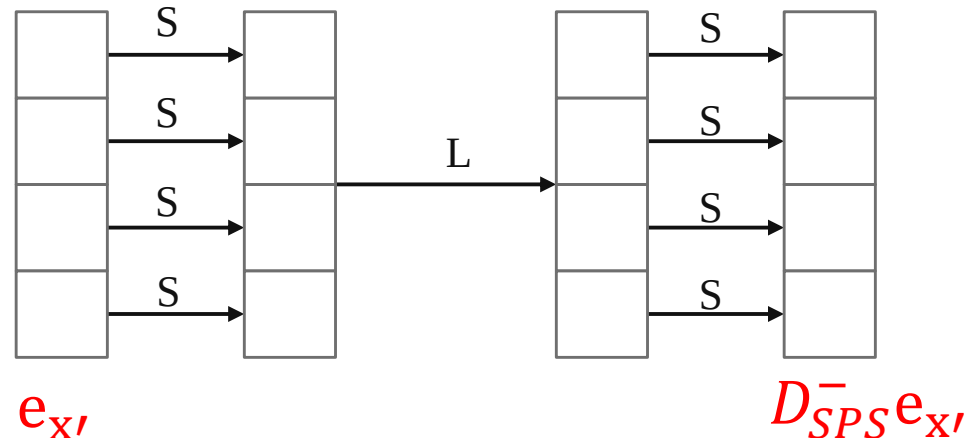
- Need to prove $\left\| D_{AES^{(8)}}^{-} e_x - U \right\|_{\infty} \leq \frac{1}{50} \cdot 2^{-128}$ for nonzero x
- Reduce to proving for 4-round AES $\left\| D_{AES^{(4)}}^{-} e_x - U \right\|_2^2 \leq \frac{1}{50} \cdot 2^{-128}$ for nonzero x
- Idea: write $AES^{(8)} = AES^{(4)} \circ AES^{(4)}$
- **Break dependency** between composition of two $AES^{(4)}$ calls by **Cauchy-Schwarz**
- Note: reduction works for any key-alternating cipher (with independent round-keys)
- Motivation: 2-norm (Euclidean norm) **easier to control** by geometry

Tracking 2-norm Contraction

- Need to prove for 4-round AES $\left\| D_{AES(4)}^- e_x - U \right\|_2^2 \leq \frac{1}{50} \cdot 2^{-128}$ for nonzero x
- Main idea: from x **track** the 2-norm of distribution through cipher operations
 - Somewhat analogous to tracking **differential characteristics**
- Goal: prove operations contract 2-norm of input vector (distribution) quickly

Main Result for Super-Sbox

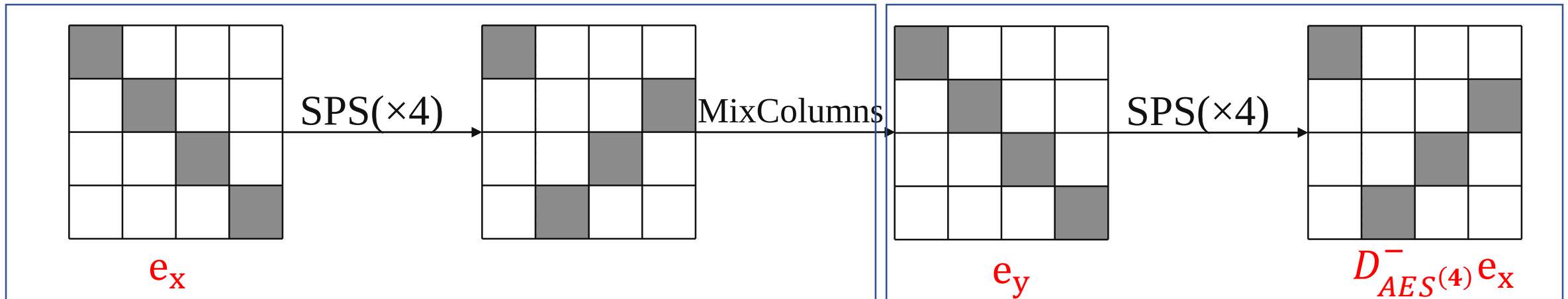
- Need to prove for 4-round AES $\left\| D_{AES(4)}^- e_x - U \right\|_2^2 \leq \frac{1}{50} \cdot 2^{-128}$ for nonzero x
- For input diff $x' \in \{0,1\}^{32}$ to SPS consider output difference distribution $D_{SPS}^- e_{x'}$
- Main technical result: for “**typical**” nonzero diff x' : $\|D_{SPS}^- e_{x'} - U\|_2^2 \lesssim \frac{1}{280} \cdot 2^{-32}$
 - Diff x' “**typical**” if it has 3 or 4 active Sboxes



Proving Main Result from Super-Sbox Analysis

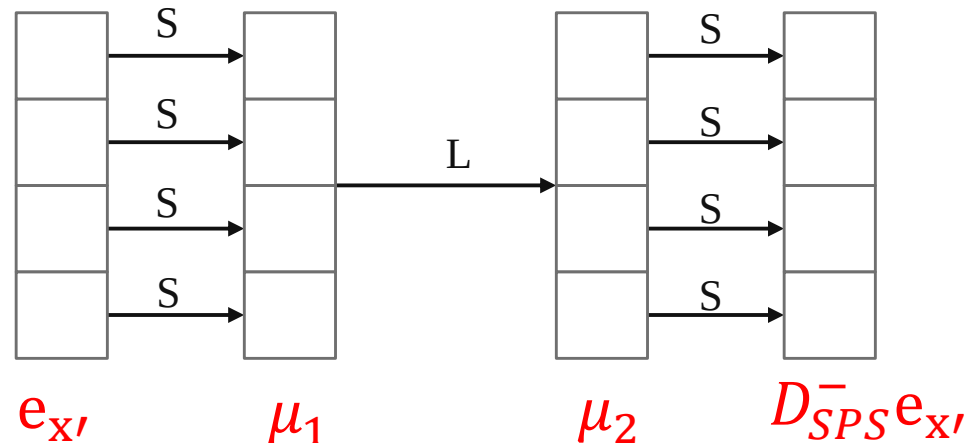
- For “**typical**” nonzero diff x' to super-Sbox: $\|D_{SPS}^- e_{x'} - U\|_2^2 \lesssim \frac{1}{280} \cdot 2^{-32}$
- Use this to prove $\left\| D_{AES(4)}^- e_x - U \right\|_2^2 \leq \frac{1}{50} \cdot 2^{-128}$:
 - With high probability, first 2 AES rounds propagate x to “**good**” difference y : each of the 4 super-Sboxes has a “**typical**” input diff
 - Assuming y is “**good**” then output 2-norm is (essentially)

$$\left\| D_{AES(4)}^- e_x \right\|_2^2 \lesssim \left(2^{-32} + \frac{1}{280} \cdot 2^{-32} \right)^4 \approx 2^{-128} + \frac{1}{70} \cdot 2^{-128}$$



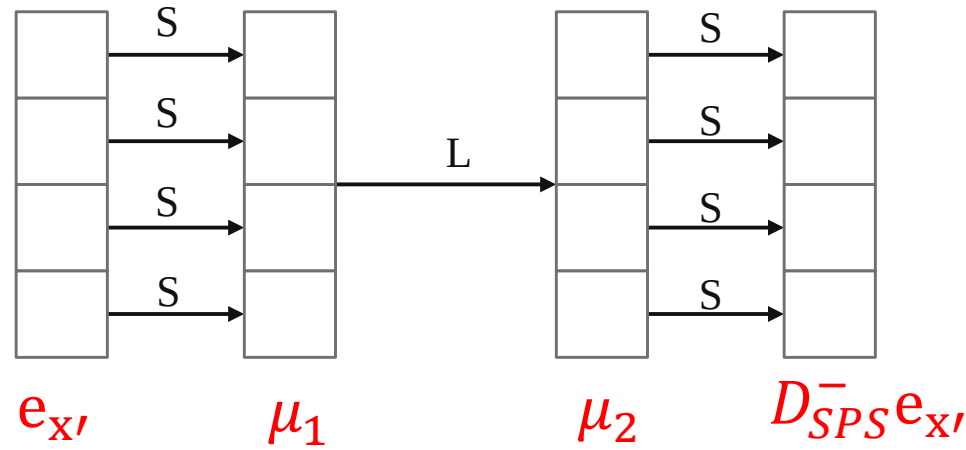
Super-Sbox Contraction - Main Technical Result

- For input diff $x' \in \{0,1\}^{32}$ to SPS consider output difference distribution $D_{SPS}^- e_{x'}$
- **Main technical result:** for “typical” nonzero diff x' : $\|D_{SPS}^- e_{x'} - U\|_2^2 \lesssim \frac{1}{280} \cdot 2^{-32}$
- Denote by μ_1, μ_2 distributions after application of 1st Sbox layer and L
- $\|\mu_1\|_2^2$ **easy** to calculate explicitly using D_S (μ_1 is **coordinate-wise** distribution)
 - $\|\mu_1\|_2^2$ depends on the **number of active Sboxes** of x'
- $\|\mu_2\|_2^2 = \|\mu_1\|_2^2$ as L is a deterministic permutation



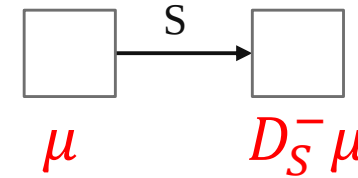
Main Technical Result

- μ_2 not simple coordinate-wise distribution
- How to analyze contraction of 2nd Sbox layer?
- Develop **Tool 1** to analyze 2-norm contraction of **Sbox layer** for **arbitrary input difference distributions**
 - Tool 1 requires information about **marginal distributions** of μ_2
- Develop **Tool 2** to analyze contraction of **MDS mapping** for **marginal distributions**



Tool 1 – 2-Norm Contraction for Sbox Layer

- Goal: analyze 2-norm contraction of **Sbox layer** for **arbitrary input difference distributions**
- Start by analyzing 2-norm contraction for **single keyed Sbox S**
- Denote by N the number of nonzero inputs to S
 - For the AES Sbox, $N = 2^8 - 1 = 255$
- Assume input difference distribution **supported on nonzero diffs**, $\mu \in \mathbb{R}^N$
- Let $U_1 \in \mathbb{R}^N$ be uniform distribution
- Fact: $\|D_S^- \mu - U_1\|_2^2 \leq \lambda_2 \|\mu - U_1\|_2^2$
- λ_2 is **2nd largest eigenvalue** of $(D_S^-)^T D_S^-$
 - D_S^- is (doubly) stochastic: $D_S^- U_1 = U_1 \rightarrow \lambda_1 = 1$
- For AES, $\lambda_2 = \left(\frac{18}{256}\right)^2 \approx \frac{1}{202}$, easily computed



Tool 1 – 2-Norm Contraction for Sbox Layer

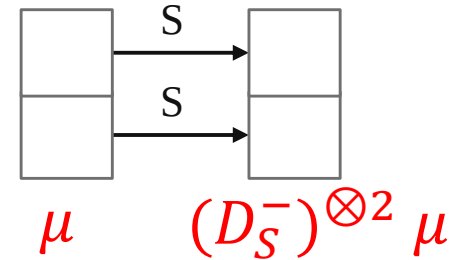
- Fact: $\|D_S^- \mu - U_1\|_2^2 \leq \lambda_2 \|\mu - U_1\|_2^2$, λ_2 is **2nd largest eigenvalue** of $(D_S^-)^T D_S^-$
- Proof:
- Let $\langle, \rangle$ be the standard inner product over $\mathbb{R}^N$: $\langle u, v \rangle = \sum_i u[i]v[i]$
- $A = (D_S^-)^T D_S^-$ is symmetric and real -> **orthogonally diagonalizable**
- Let $v_1, v_2, \dots, v_N$ be orth **eigenbasis** for A with **eigenvalues** $\lambda_1 \geq \lambda_2 \geq \dots \lambda_N \geq 0$
- D_S^- is (doubly) stochastic: $D_S^- U_1 = U_1$ -> $AU_1 = U_1$ -> $v_1 = U_1, \lambda_1 = 1$
- Write $\mu - U_1 = \sum_i \alpha_i v_i$, where $\alpha_i = \langle \mu - U_1, v_i \rangle$
- $\alpha_1 = \langle \mu - U_1, v_1 \rangle = \langle \mu - U_1, U_1 \rangle = \langle \mu, U_1 \rangle - \langle U_1, U_1 \rangle = N^{-1} - N^{-1} = 0$
 - For any distribution v , $\langle v, U_1 \rangle = \sum_i v[i]U[i] = N^{-1} \sum_i v[i] = N^{-1}$
- $\|\mu - U_1\|_2^2 = \langle \mu - U_1, \mu - U_1 \rangle = \langle \sum_{i \geq 2} \alpha_i v_i, \sum_{i \geq 2} \alpha_i v_i \rangle = \sum_{i \geq 2} (\alpha_i)^2$

Tool 1 – 2-Norm Contraction for Sbox Layer

- Fact: $\|D_S^- \mu - U_1\|_2^2 \leq \lambda_2 \|\mu - U_1\|_2^2$, λ_2 is **2nd largest eigenvalue** of $(D_S^-)^T D_S^-$
- Proof (cont.):
- $\|\mu - U_1\|_2^2 = \sum_{i \geq 2} (\alpha_i)^2$
- $\|D_S^- \mu - U_1\|_2^2 = \sum_{i \geq 2} (\alpha_i)^2 \lambda_i \leq \lambda_2 \sum_{i \geq 2} (\alpha_i)^2 = \lambda_2 \|\mu - U_1\|_2^2$

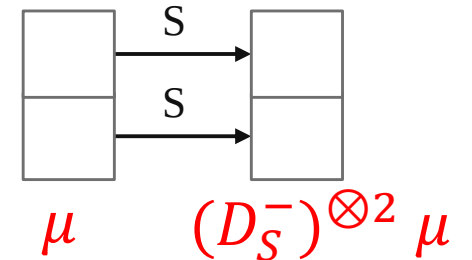
Tool 1 – 2-Norm Contraction for Sbox Layer

- Next, analyze 2-norm contraction for layer with 2 keyed Sboxes S
- Assume input diff distribution **supported on nonzero diffs to each of 2 Sboxes S** , $\mu \in \mathbb{R}^{N^2}$
- Let $U_2 \in \mathbb{R}^{N^2}$ be uniform distribution
- Denote the corresponding operator by $(D_S^-)^{\otimes 2}$ (Kronecker product)
 - Note: $(D_S^-)^{\otimes 2} U_2 = U_2$
- Want something like $\|(D_S^-)^{\otimes 2} \mu - U_2\|_2^2 \leq \lambda_2^2 \|\mu - U_2\|_2^2$
- **Not true** in general:
- If μ is product distribution on marginals: $\mu = U_1 \times \mu_2$
- As $D_S^- U_1 = U_1$, first coordinate **does not contract**
- Overall contraction $\approx \lambda_2$
 - Implies **worst-case** contraction only λ_2 (also true for general Sbox layer)
- For refined analysis, consider contraction of **marginals separately**



Tool 1 – 2-Norm Contraction for Sbox Layer

- For refined analysis, consider contraction of **marginals separately**
- $\|(D_S^-)^{\otimes 2} \mu - U_2\|_2^2 \leq \lambda_2^2 \|\mu - U_2\|_2^2$
 $+ N^{-1} \lambda_2 \|\mu^{\subseteq \{1\}} - U_1\|_2^2 + N^{-1} \lambda_2 \|\mu^{\subseteq \{2\}} - U_1\|_2^2$
 - $\mu^{\subseteq \{1\}}, \mu^{\subseteq \{2\}}$ are marginal distributions of μ on 1st and 2nd coordinates
- Proved formally by **decomposition** of $\mu - U_2$ into **3 orthogonal** components



Tool 1 – 2-Norm Contraction for Sbox Layer

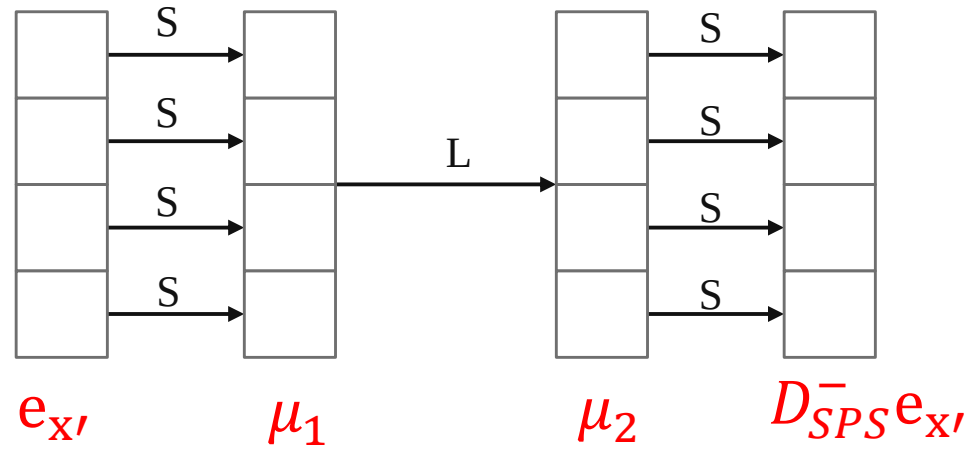
- Analyze general 2-norm contraction for layer with k keyed Sboxes S
- Assume input diff distribution **supported on nonzero diffs to each of k Sboxes S** ,
 $\mu \in \mathbb{R}^{N^k}$
- Let $U_k \in \mathbb{R}^{N^k}$ be uniform distribution
- Denote the corresponding operator by $(D_S^-)^{\otimes k}$ (Kronecker product)
 - Note: $(D_S^-)^{\otimes k} U_k = U_k$
- $\|(D_S^-)^{\otimes k} \mu - U_k\|_2^2 \leq \sum_{I \subseteq [k], I \neq \emptyset} N^{|I|-k} \lambda_2^{|I|} \|\mu^{\subseteq I} - U_{|I|}\|_2^2$
 - $\mu^{\subseteq I}$ is marginal distribution of μ on coordinates of $I \subseteq [k]$
- Note: marginals on **fewer coordinates contract less**

Tool 1 – 2-Norm Contraction for Sbox Layer

- $\|(D_S^-)^{\otimes k} \mu - U_k\|_2^2 \leq \sum_{I \subseteq [k], I \neq \emptyset} N^{|I|-k} \lambda_2^{|I|} \|\mu^{\subseteq I} - U_{|I|}\|_2^2$
- Assuming input diff distribution $\mu \in \mathbb{R}^{N^k}$ supported on nonzero diffs to each of k Sboxes S
- **Removing support assumption:** decompose μ to orthogonal distributions, each supported on subset of active Sboxes $J \subseteq [k]$
- Apply above formula to each distribution with $k' = |J|$

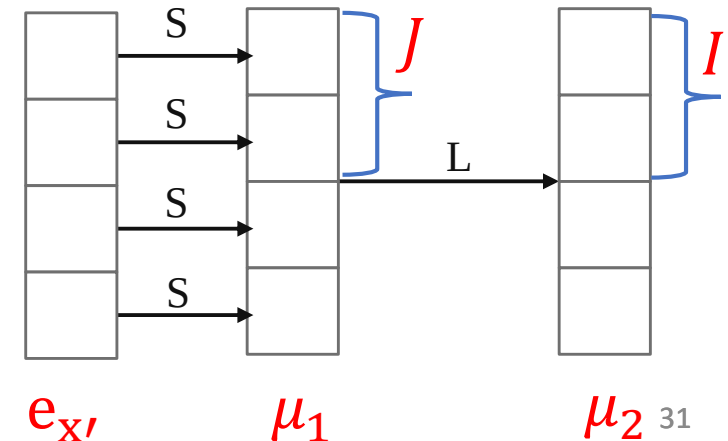
Analytic Tools

- **Tool 1** to analyzes **2**-norm contraction of **Sbox layer** for **arbitrary input difference distributions**
- Tool 1 requires information about marginal distributions of μ_2
- Develop **Tool 2** to analyze contraction of **MDS mapping** for **marginal distributions**



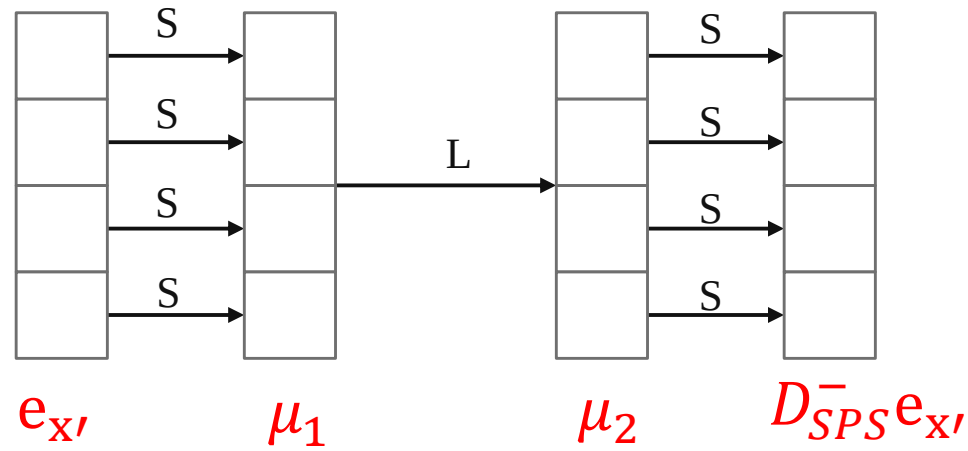
Tool 2 – 2-Norm Contraction for MDS Mapping

- Assume that μ_1 is a **product distribution on coordinates**
- Denote $\mu_2 = D_L \mu_1$
- Goal: bound $\|\mu_2^{\subseteq I}\|_2^2$ for all I
- Well-known result: MDS mappings **preserves 2-norm** for subsets of **equal size**
 - If $|I| = |J|$, then $\|\mu_2^{\subseteq I}\|_2^2 \leq \|\mu_1^{\subseteq J}\|_2^2$
- New result: If $|I| = |J|$ and $\mathbf{x}'$ has more than $|I|$ active Sboxes $\rightarrow \|\mu_2^{\subseteq I}\|_2^2 \ll \|\mu_1^{\subseteq J}\|_2^2$
 - For MDS layer, marginals on **fewer coordinates contract more**
- Main idea of proof: express μ_1 in **Fourier domain**
 - $\mu_2^{\subseteq I}$ **convolution** of (linearly transformed) marginals of μ_1



Overall Contraction of Sbox and MDS Layers

- For Sbox layer, marginals on **fewer coordinates contract less**
- For MDS layer, marginals on **fewer coordinates contract more**
- Together they contribute to the significant contraction of the super-Sbox



Extension to More Rounds

- For any nonzero different $\mathbf{x}$ to 4-round AES, $\left\| D_{AES(4)}^{-} \mathbf{e}_x - U \right\|_2^2 \leq \frac{1}{50} \cdot 2^{-128}$
- To extend to more rounds, prove **general worst-case contraction** for 4-round AES:
- Consider **arbitrary** difference distribution μ at input of 4-rounds
- Prove $\left\| D_{AES(4)}^{-} \mu - U \right\|_2^2 \leq \alpha \|\mu - U\|_2^2$ for some small α
 - Best known α is 2^{-80} by Beyne, Leander, and Schütt
- For any nonzero $\mathbf{x}$, $\left\| D_{AES(8)}^{-} \mathbf{e}_x - U \right\|_2^2 = \left\| D_{AES(4)}^{-} D_{AES(4)}^{-} \mathbf{e}_x - U \right\|_2^2 \leq \frac{1}{50} \cdot 2^{-128} \cdot \alpha$
- More generally $\left\| D_{AES(4r+4)}^{-} \mathbf{e}_x - U \right\|_2^2 \leq \frac{1}{50} \cdot 2^{-128} \cdot \alpha^r$

Conclusions and Open Problems

- We prove that any nonzero differential in 8-round AES has probability within $\frac{1}{50} \cdot 2^{-128}$ of the ideal $\frac{1}{2^{128}-1}$
- Proof uses **tools** to analyze norm contraction of **Sbox layer** and **MDS mapping**
- Tools could be useful for analyzing **additional block ciphers**
- Open problems:
 - Further **improve** results
 - Analyze differentials\2-wise independence of ciphers with **different structure than AES**

Thanks for your attention!